FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA TEST SERIES

FIITJEE

JEE (Main), 2015

CONCEPT RECAPITULATION TEST - IV

Time Allotted: 3 Hours **Maximum Marks: 360**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 30)** contains 30 multiple choice questions which have only one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION – A

Single Correct Choice Type

This section contains **30 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. An object of mass m is moving in a uniform circular motion in the xy plane. The circle has radius R and the object is moving around the circle with speed v. The motion is projected onto the x axis where it appears as simple harmonic motion according to $x(t) = R \cos(\omega t + \phi)$, here ω is equal to
   (A) v/R (B) m^2R
   (C) R/v (D) $v/(R \sin \omega t)$

2. In a standing wave on a string, the spacing between nodes is Δx. If the tension in the string is doubled but the frequency of the standing waves is fixed, then the spacing between the nodes will change to
   (A) $\sqrt{2}\,\Delta x$ (B) Δx
   (C) $\Delta x/2$ (D) $\Delta x/\sqrt{2}$

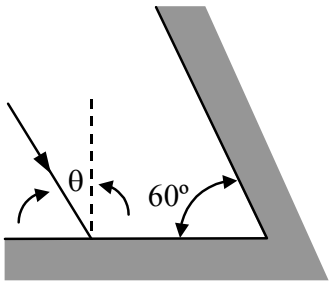
3. A 10 cm long rubber band obeys Hooke's law. When the rubber band is stretched to a total length of 12 cm the lowest resonant frequency is f_0. The rubber band is then stretched to a length of 13 cm. The lowest resonant frequency will now be
   (A) higher than f_0
   (B) the same as f_0
   (C) lower than f_0
   (D) changed, but the direction of the change depends on the elastic constant and the original tension

4. Objects A and B are initially in thermal equilibrium. Objects A and C are originally not in thermal equilibrium, but the two are placed in thermal contact and quickly reach thermal equilibrium. After doing this
   (A) B and C will also be in thermal equilibrium
   (B) B and C could be in thermal equilibrium, but might not be
   (C) B and C cannot be in thermal equilibrium
   (D) none of these

Space for Rough work

5. Four different containers each hold 0.5 moles of one of the following gases. Which gas is at the highest temperature?
(A) 8.0 L of helium gas at 120 kPa
(B) 6.0 L of neon gas at 160 kPa
(C) 4.0 L of argon gas at 250 kPa
(D) 3.0 L of krypton gas at 300 kPa

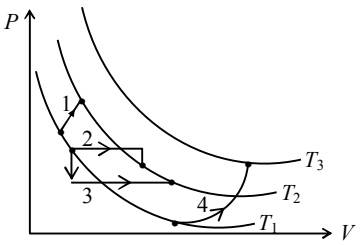
6. A ray of light strikes one of two mirrors that meet at a 60° angle as shown in figure. As the angle of incidence θ is increased, the angle between that final reflected ray and the incident ray
(A) increases
(B) decreases
(C) remains the same
(D) the question cannot be answered without solving each incident angle individually



7. Which of these statements is most correct?
(A) diffraction occurs only for transverse waves
(B) diffraction is proof that light can behave like a wave
(C) diffraction explains rainbows
(D) red sunsets are a diffraction phenomena

8. A gas mixture consists of molecules of type 1, 2 and 3 with molecular masses $m_1 > m_2 > m_3$. Which gas has maximum average translational kinetic energy?
(A) 1
(B) 2
(C) 3
(D) all has same average KE

9. The figure here shows four paths traversed by a gas on a P – V diagram. Which path has max change in internal energy of gas?
(A)1
(B)2
(C)3
(D)4



10. A carnot engine operates between the temperature T_H = 850 K and T_L = 300 K. The engine performs 1200 J of work each cycle which takes 0.25 sec. How much energy is delivered as heat to the low temperature reservoir each cycle?
(A) 655 J
(B) 1855 J
(C) 1200 J
(D) 600 J

Space for Rough work

11. Of the following quantities, which one has dimensions different from the remaining three
(A) energy per unit volume
(B) force per unit area
(C) product of voltage and charge per unit volume
(D) angular momentum per unit mass

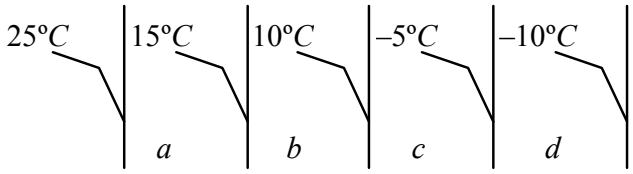
12. The resistance R = V/i where V = 100 $\pm$ 5 volts and i = 10 $\pm$ 0.2 amperes. What is the total error in R
(A) 5% (B) 7%
(C) 5.2% (D) 5/2%

13. A ray of light travels from an optically denser to rarer medium. The critical angle for the two media is C. The maximum possible deviation of the ray will be
(A) $\left(\frac{\pi}{2} - C\right)$ (B) 2C
(C) $\pi - 2C$ (D) $\pi - C$

14. 150, 225, 300, 375, is a series of resonant frequency. Which frequency is missing lower than $400H_z$
(A) 250 (B) 350
(C) 75 (D) 200

15. The following four waves are sent along four wires with the same linear mass densities. Which string has maximum tension
(A) $y_1 = 3 \sin (9x - 3t)$ (B) $y_2 = 6 \sin (2x - t)$
(C) $y_3 = 1 \sin (4x - t)$ (D) $y_4 = 2 \sin (x - 2t)$

16. The figure shows the face and interface temperature of a composite slab consisting of four materials, of identical thickness, through which the heat transfer is steady. Which material has maximum thermal conductivity
(A) a
(B) b
(C) c
(D) d



Space for Rough work

17. The wavelength of radiation emitted is λ_0, when an electron jumps from the third to the second orbit of hydrogen atom. For the electron jump from the fourth to second orbit of the hydrogen atom, the wavelength of radiation emitted will be

(A) $\frac{16}{25}\lambda_0$ (B) $\frac{20}{27}\lambda_0$

(C) $\frac{27}{20}\lambda_0$ (D) $\frac{25}{16}\lambda_0$

18. A radioactive substance disintegrates as follows ${}_{92}X^{234} \rightarrow {}_{87}Y^{222}$. The number of α and β-particles that are emitted in this process are

(A) 3, 5 (B) 5, 3

(C) 3, 3 (D) 3, 1

19. Suppose the daughter nucleus in a nuclear decay is itself radioactive. If λ_d and λ_m denote the decay constants of daughter and mother nuclei and N_d and N_m the numbers of daughter and mother nuclei present at a time, then the number of daughter nuclei becomes constant when

(A) $\lambda_m N_m = \lambda_d N_d$ (B) $\lambda_m N_d = \lambda_d N_m$

(C) $N_m - N_d = \lambda_m - \lambda_d$ (D) $N_m + N_d = \lambda_m + \lambda_d$

20. Consider the fission reaction, ${}_{92}U^{236} \rightarrow X^{117} + Y^{117} + n + n$ two nuclei of same mass number 117 are found plus two neutrons. The binding energy per nucleon of X and Y is 8.5 MeV whereas of U^{236} is 7.6 MeV. The total energy liberated will be about

(A) 2 MeV (B) 20 MeV

(C) 200 MeV (D) 2000 MeV

21. Energy released in the fission of a single ${}_{92}U^{235}$ nucleus is 200 MeV. The fission rate of ${}_{92}U^{235}$ fuelled reactor operating at a power level of 5 watt is

(A) 1.56×10^{10}/sec (B) 1.56×10^{11}/sec

(C) 1.56×10^{20}/sec (D) 1.56×10^{-17}/sec

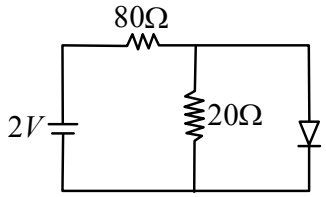
22. In the circuit shown, the current through the ideal diode is

(A) 75 mA

(B) 20 mA

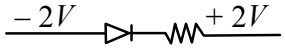
(C) 100 mA

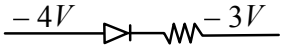
(D) 25 mA

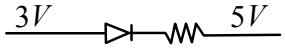


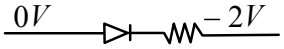
Space for Rough work

23. The forward biased diode is

(A) 

(B) 

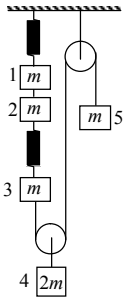
(C) 

(D) 

24. At $t = 0$, a particle is at rest at $x = 0$ where x is the position of the particle with respect to origin. The acceleration of the particle depends on its position as $a = \frac{3x^2}{2}$. The velocity of the particle at t = 1s will be

(A) 8 m/s
(B) 4 m/s
(C) 2 m/s
(D) 0

25. All the strings, springs and pulleys shown in figure are ideal. Initially the system is in equilibrium and blocks are at rest. Now, the upper spring is cut. The acceleration of block 1 just after cutting the upper spring will be



(A) $\frac{g}{3}$
(B) $\frac{g}{2}$
(C) 2g
(D) $\frac{2g}{3}$

26. A water tank having a uniform cross-sectional area A has an orifice of area of cross-section a at the bottom through which water is discharging. There is constant inflow of Q into the tank through another pipe. If $a\sqrt{2g} = K$, the time taken for the water level to rise from h_1 to h_2 will be

(A) $t = \frac{A}{K^2}\left[K\left(\sqrt{h_1} - \sqrt{h_2}\right) - Q\ln\left(\frac{Q - K\sqrt{h_2}}{Q - K\sqrt{h_1}}\right)\right]$

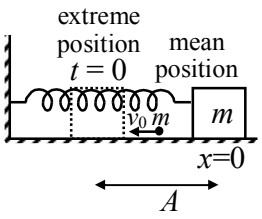
(B) $t = \frac{3A}{K^2}\left[K\left(\sqrt{h_1} - \sqrt{h_2}\right) - Q\ln\left(\frac{Q - K\sqrt{h_2}}{Q - K\sqrt{h_1}}\right)\right]$

(C) $t = \frac{2A}{K^2}\left[K\left(\sqrt{h_1} - \sqrt{h_2}\right) - Q\ln\left(\frac{Q - K\sqrt{h_2}}{Q - K\sqrt{h_1}}\right)\right]$

(D) $t = \frac{A}{2K^2}\left[K\left(\sqrt{h_1} - \sqrt{h_2}\right) - Q\ln\left(\frac{Q - K\sqrt{h_2}}{Q - K\sqrt{h_1}}\right)\right]$

Space for Rough work

27. A block of mass m is performing SHM with an amplitude A, on a frictionless surface. When it is at extreme position, a bullet of mass m moving with a velocity v_0 collides and gets embedded into it at a time t = 0 ($mv_0^2 = 2kA^2$) then the displacement x at any time t will be



(A) $x = -\sqrt{2}A\sin\left(\sqrt{\frac{k}{2m}}t + \pi/4\right)$

(B) $x = -\sqrt{2}A\sin\sqrt{\frac{k}{2m}}t$

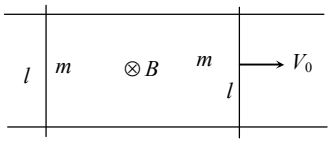
(C) $x = -\sqrt{2}A\sin\left(\sqrt{\frac{k}{2m}}t + \frac{\pi}{3}\right)$

(D) $x = -A\sin\sqrt{\frac{k}{m}}t$

28. An electron of KE 100eV collides with stationary helium ion He^+ in its ground state and excites it to a higher energy level. After collision He^+ ion emits two photons in succession with wavelength 1085Å and 304Å. Then the KE of electron after collision will be (Neglect the recoil speed of He^+)

(A) 47.7 eV
(B) 94.4 eV
(C) 20.2 eV
(D) 10.1 eV

29. Two parallel conducting rails, separated by a distance l, are placed horizontally in a region of uniform magnetic field $\vec{B}$, perpendicular to the plane of the rails as shown in the figure. Two conducting wires, each of length l are placed so as to slide on parallel conducting rails without friction. One of the wires is given a velocity v_0 parallel to the rails. Loss in kinetic energy of the system till the steady state is achieved will be



(A) Zero
(B) $\frac{3}{4}mv_0^2$
(C) $\frac{3}{8}mv_0^2$
(D) $\frac{1}{4}mv_0^2$

30. A round metal hoop is suspended on the edge by a hook. The hoop can oscillate side to side in the plane of the hoop, or it can oscillate back and forth in a direction perpendicular to the plane of the hoop. For which mode will the frequency of oscillation be larger?

(A) oscillations in the plane of the hoop
(B) oscillations perpendicular to the plane of the hoop
(C) the frequency of oscillation will be the same in either mode
(D) none of these

Space for Rough work

Chemistry

PART – II

SECTION – A
Single Correct Choice Type

This section contains **30 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. Minerals kinoite is a silicate anion having chain of three SiO_4 tetrahedra. If in kinoite the three SiO_4 tetrahedra shares corners with adjacent tetrahedra then the overall charge on silicate anion is:
(A) – 4 (B) – 8
(C) – 6 (D) – 2

2. When compounds

$Cu(NO_3)_2$	$Hg(NO_3)_2$	$NaNO_3$	$AgNO_3$
I	II	III	IV

are heated (upto 500°C) seperately then a brown gas is formed in case of
(A) I and II (B) II and III
(C) II, III and IV (D) I, II and IV

3. Consider the following order:
(I) Thermal stability: $BeSO_4 < MgSO_4 < CaSO_4 < SrSO_4 < BaSO_4$.
(II) Bond angle: $OCl_2 > SF_2 > ASH_3 > H_2Se$
(III) Bond order of O_2^+, NO and N_2^- is 2.5.
(IV) NCl_3 on hydrolysis gives HOCl but BCl_3 gives HCl.
Identify which of the above four statements are correct.
(A) I, II, III and IV (B) II, III and IV
(C) I, II and IV (D) I and II

4. Identify the following statements as True (indicated by T) or False (indicated by F)
(I) Dipole moment order of $NH_3 > SbH_3 > AsH_3 > PH_3$
(II) F_2 on reaction with water produces HF, O_2 and O_3
(III) Number of S—S bonds in $H_2S_nO_6$ are n–2
(IV) PH_3 can be produced by hydrolysis of Ca_3P_2
(A) TFTT (B) TTFF
(C) TFFT (D) TTFT

5. Identify the incorrect statement for XeF_6
(A) XeF_6 reacts with excess H_2O to form XeO_3.
(B) XeF_6 may act as a fluoride donor during reaction with group I element.
(C) XeF_6 on reaction with small amount of H_2O form $XeOF_4$
(D) None of the above

Space for Rough work

6. For complex $[Co(NH_3)_5CO_3]ClO_3$, what is the spin only magnetic moment, oxidation number coordination number and d-electrons on transition metal.
(A) $\sqrt{3}$ BM, 0, 7, 2
(B) $\sqrt{15}$ BM, 3, 6, 0
(C) 0 BM, 3, 6, 6
(D) $\sqrt{15}$ MB, 2, 7, 3

7. Identify the incorrect statement
(A) $HgCl_2$ is prepared by heating mercury in chlorine.
(B) HgO decomposes on heating whereas Al_2O_3 has high thermal stability
(C) Mac-Aurthor's process is used to extract platinum
(D) NO_2 undergoes disproportionation reaction under alkaline medium.

8. Identify the incorrect:
(A) $Ni^{+2}(aq) + \text{dimethylglyoxime}(aq) + NH_3(aq) \longrightarrow \text{Red ppt/colour}$
(B) $Hg^{2+}(aq) + SCN^-(aq) + CO^{2+}(aq) \longrightarrow \text{Blue ppt/colour}$
(C) When $Mn(NO_3)_2$ (aq) is treated with sodium bismuthate in presence of concentration HNO_3 a purple coloured solution is obtained.
(D) None of these

9. Which of the following have same number of p-electrons as the 4d-electrons of Palladium?
(A) Si
(B) S
(C) P
(D) None of these

10. If the longest wavelength in Balmer series of Li^{2+} is 'x' then the value of shortest wavelength of He^+ Paschen series will be
(A) $\frac{36x}{5}$
(B) $\frac{20x}{36}$
(C) $\frac{45x}{16}$
(D) $\frac{36x}{20}$

11. The ratio of rms velocity of N_2, average velocity of H_2 and most probable velocity of CO at 27°C is
(A) $\sqrt{\frac{3}{28}} : \sqrt{\frac{4}{\pi}} : \sqrt{\frac{1}{14}}$
(B) 1.224 : 1.128 : 1
(C) 0.33 : 1.113 : 0.27
(D) Both (A) and (C)

12. Element 'X' having atomic mass equal to 109.75 g crystallises in face centred cubic (fcc) structure. If the density of crystal is 9.0 g/cm^3 then the edge length of the unit is
(A) 450 pm
(B) 750 pm
(C) 900 pm
(D) 600 pm

Space for Rough work

13. In solubility of hydroxide $A(OH)_2$ in a buffer of pH = 11 is found to be 0.0416 g/L. Molar solubility of $A(OH)_2$ in pure water would be (in mol/lit). Use molecular weight of $A(OH)_2$ as 104 g/mol
(A) 10^{-3} (B) 2.16×10^{-3}
(C) 10^{-4} (D) 2.16×10^{-4}

14. Consider following equilibrium at 400 K and 50 atm

$$3A \rightleftharpoons 2B + 2C + D$$

If the equilibrium pressure of A is 5 atm then K_c will be:
(A) 38.32 (B) 19.16
(C) 9.58 (D) 28.74

15. For adiabatic reversible expansion, initial pressure and temperature are respectively 1 atm and 27°C. If final pressure is 0.25 atm then final temperature will be: (Assume the gas is monoatomic)
(A) –100°C (B) –50°C
(C) –25°C (D) 0°C

16. For the combustion of C_6H_{10} with required amount of oxygen. Which of the following is correct? {consider the state of each reactant and product same as at room temperature}.
(A) $\Delta H > \Delta E$ (B) $\Delta H = \Delta E$
(C) $\Delta H < \Delta E$ (D) None of these

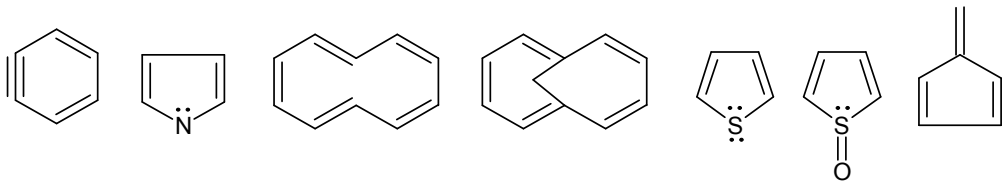
17. 10 g sample of compound 'X' undergoes a first order reaction. If $t_{1/2}$ for this reaction is 20 min then the time needed to consume T_g of sample is
(A) 40 min (B) 32 min
(C) 34.75 min (D) 38 min

18. Identify the incorrect statement regarding adsorption of gas on solid surface.
(A) Adsorption can be endothermic
(B) Physisorption may get transformed into a Chemisorption at high temperature.
(C) Chemisorption is very slow as compared to physisorption because chemisorption have higher activation energy.
(D) Chemisorption is more exothermic than physisorption

19. Elevation in boiling point of aqueous glucose solution is 0.104 K. Osmotic pressure for same aqueous glucose solution was found to be 7.38 atm at 300 K. The density of aqueous glucose solution is:
Take R = 0.082 L. atm. Mol^{-1} K^{-1} and K_b = 0.52 K Kg mol^{-1}
(A) 1.036 g mL^{-1} (B) 1.054 g mL^{-1}
(C) 1.00 g mL^{-1} (D) None of these

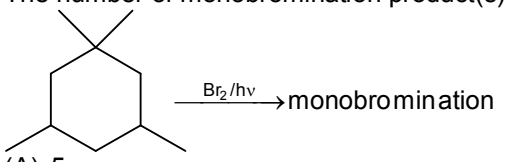
Space for Rough work

20. Identify the total number or aromatic compound from following:



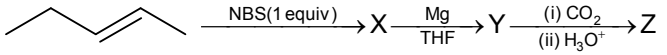
(A) Six
(B) Five
(C) Four
(D) Three

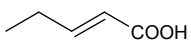
21. The number of monobromination product(s) in the following reaction is:

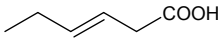


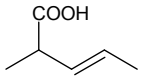
(A) 5
(B) 7
(C) 6
(D) 8

22. Identify the major produt (Z) in the following reaction sequence.



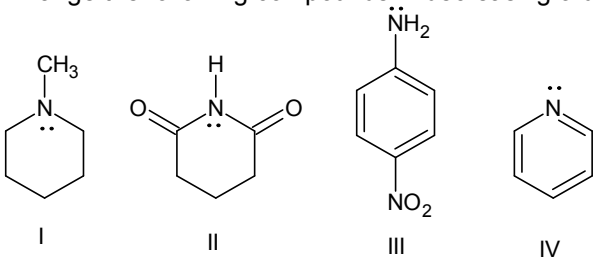
(A) 

(B) 

(C) 

(D) 

23. Arrange the following compounds in decreasing order of their basicity:

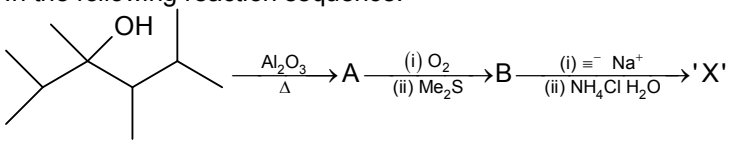


I II III IV

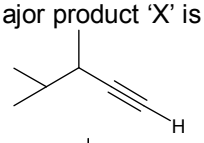
(A) I > IV > II > III
(B) I > III > IV > II
(C) I > IV > III > II
(D) I > III > II > IV

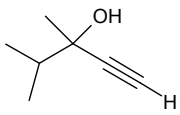
Space for Rough work

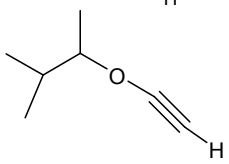
24. In the following reaction sequence:



The major product 'X' is

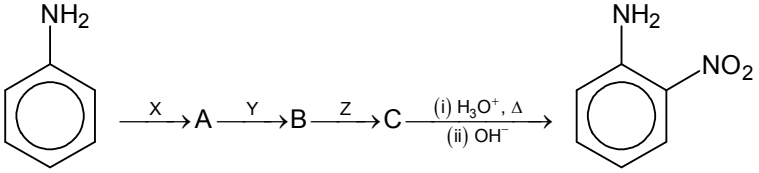
(A) 

(B) 

(C) 

(D) Both B and C

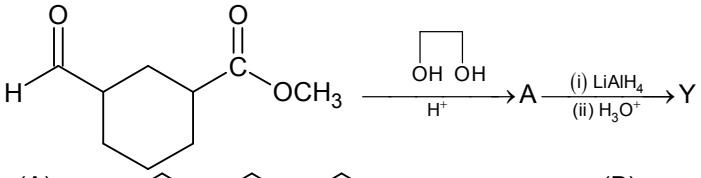
25. For the following reaction sequence

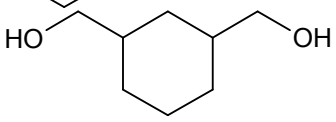


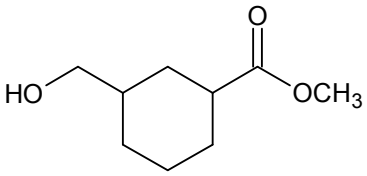
Identify the correct sequence of reagents X, Y and Z

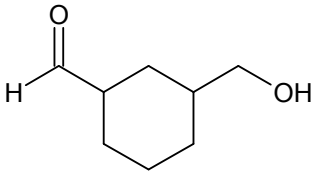
	X	Y	Z
(A)	Conc. HNO_3	CH_3COCl/base	Conc. H_2SO_4
(B)	CH_3COCl, Base	Conc. HNO_3	Conc. H_2SO_4
(C)	Conc. H_2SO_4	CH_3COCl, base	Conc. HNO_3
(D)	CH_3COCl, base	Conc. H_2SO_4	Conc. HNO_3

26. In the following reaction identify the major product 'X':



(A) 

(B) 

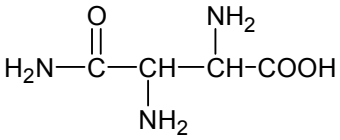
(C) 

(D) Both (B) and (C)

Space for Rough work

27. Identify incorrect statement regarding Hunsdieker reaction:
(A) It can be given by silver salt of carboxylic acid
(B) Major product of reaction is halide of corresponding carboxylic acid
(C) It follows radical mechanism
(D) None of the above

28. What will be the net change on following amino acid at pH = 3



(A) +2
(B) +3
(C) –1
(D) Zero

29. When C_4H_{10} undergoes chlorination by reaction with $Cl_2(g)$ in presence of ultraviolet light then how many stereoisomers are possible for dichloroderivatives.
(A) Two
(B) Three
(C) Four
(D) Five

30. The expression of K_p for the following reaction will be:

$$KMnO_4(aq) + H_2C_2O_4(aq) + HCl(aq) \rightleftharpoons MnO(s) + CO_2(g) + H_2O(\ell) + KCl(aq)$$

P_{CO_2} is partial pressure of carbon dioxide

(A) $Kp = P_{CO_2}$
(B) $K_p = \left(P_{CO_2}\right)^2$
(C) $Kp = \left(P_{CO_2}\right)^5$
(D) $K_P = \left(P_{CO_2}\right)^{10}$

Space for Rough work

Mathematics

PART – III

SECTION – A

Single Correct Choice Type

This section contains **30 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. $\lim_{x\to 0}\frac{\tan x\cos(\sin x)-\sin(\sin x)}{\cos(\tan x)\tan x-\sin(\tan x)}$ is

   (A) $-\frac{1}{2}$ (B) –1

   (C) $\frac{1}{2}$ (D) 1

2. If $f\left(\sqrt{3-x}\right)=\frac{1}{\sqrt{x}}$ is true for exhaustive domain of 'f', then domain of f(x) is

   (A) $\left(-\sqrt{3},\ \sqrt{3}\right)$ (B) $\left[0,\ \sqrt{3}\right]$

   (C) $\left(0,\ \sqrt{3}\right]$ (D) $\left[0,\ \sqrt{3}\right)$

3. If α is a root of $z^5+z^3+z+3=0$ then

   (A) $|\alpha|\geq 1$ (B) $|\alpha|<1$

   (C) α lies on or outside the circle $|z|=\frac{1}{2}$ (D) α lies inside the unit circle $|z|=1$

4. In a quadrilateral ABCD with sides a, b, c, d the range of $\frac{a^2+b^2+c^2}{d^2}$ is

   (A) $\left(\frac{1}{3},\infty\right)$ (B) $\left(\frac{2}{3},\infty\right)$

   (C) $(1,\infty)$ (D) $(0,\infty)$

5. If a_i is negative or positive according as i is odd or even and equation $x^4+a_1x^3+a_2x^2+a_3x+5=0$ has four real roots then minimum values of a_1a_3 is equal to

   (A) 20 (B) 40

   (C) 80 (D) 160

Space for rough work

6. Let a function f(x) has the property $f(x+2)=\frac{f(x)-5}{f(x)-3}$, then which of the following must equal to f(2014)
(A) f(2004) (B) f(2010)
(C) f(2006) (D) f(2018)

7. $\lim\limits_{x\to n^-}\frac{e^{\{x\}}-\{x\}-1}{\{x\}^2}$ (where n ∈ I), is
(A) e (B) $e-\frac{1}{2}$
(C) e – 2 (D) none of these

8. From (1, 2), tangents are drawn to the curve $y^2 - 2x^3 - 4y + 8 = 0$. Then
(A) sum of x–coordinates of points of contact is zero
(B) sum of x–coordinates of points of contact is 6
(C) sum of y–coordinates of points of contact is zero
(D) sum of y–coordinates of points of contact is 4

9. If $\left|\int_a^b f(x)dx\right| = \int_a^b |f(x)|dx$ $(a \neq b)$ also f(x) ≠ 0 for any x ∈ (a, b) and $g(x)=\int_0^x f(x)dx$, then $\int_a^b f(x)g(x)dx$ when a and b are positive
(A) cannot be positive (B) cannot be negative
(C) can be less than –1 (D) nothing can be said

10. Three straight lines are drawn through a point 'P' lying in the interior of the ΔABC and parallel to its sides. The areas of the three resulting triangles with P as the vertex are s_1, s_2 and s_3 then area of ABC is equal to
(A) $\left(\sqrt{s_1}+\sqrt{s_2}+\sqrt{s_3}\right)^2$ (B) $(s_1 + s_2 + s_3)$
(C) $\sqrt{s_1s_2+s_2s_3+s_3s_1}$ (D) $s_1 + s_2 + s_3 + 2\left(\sqrt{s_1s_2}+\sqrt{s_2s_3}\right)$

11. Let a real valued differentiable function f(x) be strictly monotonic on [a, b], then
(A) $\int_a^b\left[(f(x))^2-(f(b))^2\right]dx=\int_{f(a)}^{f(b)}2x\left(b-f^{-1}(x)\right)dx$
(B) $\int_a^b\left[(f(x))^2-(f(a))^2\right]dx=\int_{f(a)}^{f(b)}2x\left(b-f^{-1}(x)\right)dx$
(C) $\int_a^b\left[f(x)-f(a)\right]^2dx=\int_{f(a)}^{f(b)}x\left(b-f^{-1}(x)\right)^2dx$
(D) none of these

Space for rough work

12. If $z_0 = \frac{1+i}{2}$, then the value of the product $(1+z_0)\left(1+z_0^2\right)\left(1+z_0^{2^2}\right).....\left(1+z_0^{2^n}\right)$ must be

(A) $2^{2^{n+1}}$ (B) $(1+i)\left(1-\frac{1}{2^{2^n}}\right)$, if $n \geq 1$

(C) $\frac{5}{4}(1+i)$, if $n = 1$ (D) 0

13. Let N be any four digit number say $x_1\, x_2\, x_3\, x_4$. Then maximum value of $\frac{N}{x_1 + x_2 + x_3 + x_4}$ is equal to

(A) 1000 (B) $\frac{1111}{4}$

(C) 800 (D) none of these

14. $\int \frac{e^{\cot x}}{\sin^2 x}$ [2 ln cosec x + sin 2x] dx is equal to

(A) $-2e^{\cot x}$ ln (cosec x) + c (B) $e^{\cot x}$ ln x + c

(C) $e^{\cot x}$ (ln (cosec x) + c (D) none of these

15. Let P be any moving point on the circle $x^2 + y^2 - 2x = 1$. AB be the chord of contact of this point w.r.t the circle $x^2 + y^2 - 2x = 0$. The locus of the circumcentre of the triangle CAB, (C being centre of the circles) is

(A) $2x^2 + 2y^2 - 4x + 1 = 0$ (B) $x^2 + y^2 - 4x + 2 = 0$

(C) $x^2 + y^2 - 4x + 1 = 0$ (D) $2x^2 + 2y^2 - 4x + 3 = 0$

16. ABCD is a quadrilateral with side lengths AB = 4, BC = 10, CD = 6 and AD = 6, and diagonal BD = 8 units. If the incircles of triangles ABD and BCD touch BD at P and Q respectively, then area of quadrilateral C_1PC_2Q (where C_1 and C_2 are incentres of triangle ABD and BCD respectively), is

(A) $3 + \frac{\sqrt{15}}{2}$ sq. units (B) 3 sq. units

(C) $\frac{\sqrt{15}}{6}$ sq. units (D) none of these

Space for rough work

17. Let S = {1, 2, 3, ….. n} be a set and two numbers x and y are drawn with replacement from S. Let P_k be the probability that $x^2 - y^2$ is divisible by k and [x] denotes greatest integer less than or equal to x. The value of P_2 is equal to

(A) $\frac{1}{2}$ (B) $\left(\frac{\left[\frac{n}{2}\right]}{n}\right)^2$

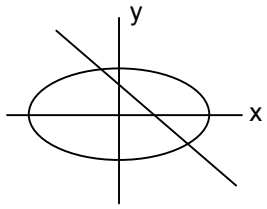
(C) $1-2.\frac{\left[\frac{n}{2}\right]}{n}+2\left(\frac{\left[\frac{n}{2}\right]}{n}\right)^2$ (D) $1-\frac{2-\left[\frac{n}{2}\right]}{n}-2\left(\frac{\left[\frac{n}{2}\right]}{n}\right)^2$

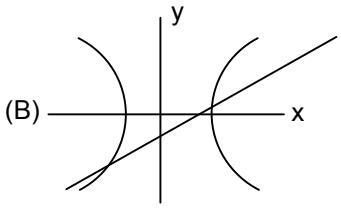
18. A eccentric person starts writing numbers from 1 to n in a row such that i^{th} number is written i^2 times. The 500th digit from the starting in such a number is
(A) 0 (B) 1
(C) 2 (D) 3

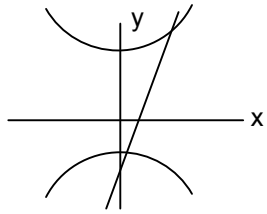
19. The value of a for which $4^{\sin x} + a2^{\sin x} - 1 \le 0 \ \forall\ x \in R$ is
(A) $a \ge -\frac{3}{2}$ (B) $a \le -\frac{3}{2}$
(C) $a \le 2$ (D) No such value exists

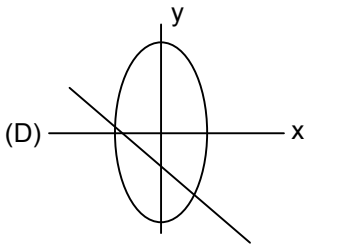
20. The number of tangents which can be drawn to the ellipse $16x^2 + 25y^2 = 400$, such that sum of perpendicular distance from the foci to any tangent is 8, is
(A) 1 (B) 2
(C) 4 (D) none of these

21. Let $a, b \in R$ where $ab \ne 0$, then the graph of the straight line $ax - y + b = 0$ and the conic section $bx^2 + ay^2 = ab$ can be

(A) 

(B) 

(C) 

(D) 

Space for rough work

22. The differential equation of a curve, passing through $\left(0, -\frac{\pi}{4}\right)$ and $(t, 0)$ is $\cos y\left(\frac{dy}{dx} + e^{-x}\right) + \sin y\left(e^{-x} - \frac{dy}{dx}\right) = e^{e^{-x}}$, the value of $t.e^{e^{-t}}$ is

(A) –1 (B) 1
(C) 2 (D) –2

23. If $(1, 2, p)$, $(2, 2p, -6)$ and $(\alpha^2 - 2\alpha, 1, 1)$ are the ordered triplets of form (x, y, z) which satisfies all the equations. $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$, $\frac{x}{b}+\frac{y}{c}+\frac{z}{a}=1$ and $\frac{x}{c}+\frac{y}{a}+\frac{z}{b}=1$ then α can be

(A) 2 (B) –2
(C) –4 (D) 6

24. If $f(x)$ is a continuous and differentiable function such that $f(t_k) = 1$ for $k = 1, 2, ...n$, where $t_k = \sum_{r=1}^{k} \frac{1}{r(r+1)(r+2)}$ then

(A) $f\left(\frac{1}{4}\right) = 1$ (B) $f\left(\frac{1}{2}\right) = 1$
(C) $f'\left(\frac{1}{2}\right) = 0$ (D) $f'\left(\frac{1}{4}\right) = 1$

25. The sum of the real values that the function $f(x) = \frac{x^2 - 3x + 2}{x^2 + x - 6}$ can not take is

(A) 2 (B) – 3
(C) $\frac{6}{5}$ (D) $\frac{4}{5}$

26. Let z_1, z_2, z_3 be the coordinates of vertices A, B, C of the triangle ABC. If $W_1 = z_1 - z_2$, $W_2 = z_3 - z_2$ and $\angle A = 90°$, then

(A) $R_e\left(\frac{W_2}{W_1}\right) = 2$ (B) $R_e(W_1\bar{W}_2) = |W_1|^2$
(C) $R_e(W_1\bar{W}_2) = |W_2|^2$ (D) $R_e\left(\frac{W_1}{W_2}\right) = 1$

Space for rough work

27. Let $\cos\alpha$ is a root of the equation $169x^2 - 26x - 35 = 0$, $-1 < x < 0$, then $\sin 2\alpha$ is

(A) $\frac{144}{169}$ (B) $-\frac{144}{169}$

(C) $\frac{120}{169}$ (D) $-\frac{131}{169}$

28. Let $(1+x+x^2)^n = a_0 + a_1x + a_2x^2 \ldots\ldots a_{2n}x^{2n}$, where n is odd integer

$S_1 = a_0 + a_4 + a_8 + \ldots\ldots$

$S_2 = a_1 + a_5 + a_9 + \ldots\ldots$

$S_3 = a_2 + a_6 + a_{10} + \ldots\ldots$

$S_4 = a_3 + a_7 + a_{11} + \ldots\ldots$

(A) $S_1 = 2S_3$ (B) $S_2 = S_4$

(C) $S_2 + S_4 = 0$ (D) $S_1 = S_2 = S_3$ or $S_1 = S_3 = S_4$

29. Let $f(n)$ be the sum of first n terms of sequence 0, 1, 1, 2, 2, 3, 3, 4, 4 ….., then

(A) $f(n) = \frac{n^2}{6}$ {n is even number)

(B) $f(n) = \frac{n^2-1}{6}$ {n is odd number)

(C) $f(n+m) - f(n-m) = nm$, where $n, m \in I^+ (n > m)$

(D) $f(n+m) - f(n-m) = \frac{4nm+1}{1}$, where $n, m \in I^+ (n > m)$

30. If $\vec{A}, \vec{B}, \vec{C}$ and $\vec{D}$ are four non-zero vectors in the same plane no two of which are collinear then which of the following hold(s) good?

(A) $(\vec{A} \times \vec{B}).(\vec{C} \times \vec{D}) = 0$ (B) $(\vec{A} \times \vec{C}).(\vec{B} \times \vec{D}) = 0$

(C) $(\vec{A} \times \vec{B}) \times (\vec{C} \times \vec{D}) = \vec{0}$ (D) $(\vec{A} \times \vec{C}) \times (\vec{B} \times \vec{D}) \neq \vec{0}$

Space for rough work